

ECS Ecosystem Governance Framework (ECS-EGF)

Field	Value
Document Name	ECS Ecosystem Governance Framework (ECS-EGF)
Version	DRAFT 0.1
Status	Draft for co-authoring by Founding Members
Governs	The ECS ecosystem: the four Essential Credential Schemas (Service, Organization, Persona, UserAgent) and every Participant in their permission trees
Governed By	The Verana Council Association (in formation), acting as the ecosystem governance authority (EGA) of the ECS ecosystem
Approved By	Pending — Incorporation General Assembly, target Q4 2026
Approval Date	Pending
License	CC-BY-SA 4.0
Anchoring	Git history and tagged releases in verana-labs/verana-council-gov plus a public URL now; on-chain GovernanceFrameworkDocument (url + digest_sri) on the ECS Ecosystem entry once the network is live

Status: DRAFT 0.1. This is a first draft published for co-authoring by the Founding Members of the Verana Council Association (in formation). It is scheduled for ratification at the Q4 2026 Incorporation General Assembly. Where this document states protocol facts, the [Verifiable Trust specification](#) and the [Verifiable Public Registry \(VPR\) specification](#) prevail. Where it states entity facts, the Council's public record and its published entity definitions prevail. During the pre-incorporation period, 2060 OÜ acts as transitional steward; every steward power carries an event-triggered sunset (see Chapter 14).

1. Introduction

This chapter is non-normative.

1.1 Why Essential Credential Schemas exist

Verifiable Trust inverts the internet's default: verify first, then connect. Before any connection is accepted, a relying party must be able to answer, by independent cryptographic verification, four questions about a counterpart:

1. **What service is this?** — answered by a Service credential.
2. **Who is accountable for it?** — answered by an Organization credential (a legal entity) or a Persona credential (an identified individual controller).
3. **What software is the user running?** — answered by a UserAgent credential.
4. **Under which governance framework is all of the above accredited?** — answered by trust resolution terminating at an Ecosystem DID and its published, digest-anchored governance framework.

The four **Essential Credential Schemas (ECS)** — Service, Organization, Persona, UserAgent — are the minimum schema set that makes those four questions resolvable for *any* Verifiable Service (VS) or Verifiable User Agent (VUA), regardless of sector. The Verifiable Trust specification defines their content ([ECS-SERVICE], [ECS-ORG], [ECS-PERSONA], [ECS-UA]) and requires that an ECS Ecosystem provide all four as CredentialSchema entries in a Verifiable Public Registry (VPR).

1.2 The ECS ecosystem and its trust-root status

This document is the ecosystem governance framework (EGF) of the **ECS ecosystem** operated by the Verana Council Association: the ecosystem that publishes the four ECS on the Verana network and accredits the Participants who issue and verify credentials under them.

The ECS ecosystem holds no protocol privilege. Per the Verifiable Trust specification's whitelist mechanism ([WL]), each VS and VUA operator maintains its own list of trusted ECS Ecosystem DIDs. The Council's ECS ecosystem becomes a trust root **only** because operators choose to recognize it — a client-side, revocable act of recognition, not a protocol grant. This document is written so that such recognition remains rational: public rules, non-discretionary accreditation, symmetric records, and enforcement bound to on-chain state.

1.3 Position in the document architecture

The Verana Council governs three framework documents:

- the **Network Governance Framework (Network GF)** — the constitutional layer that every EGF anchored on the Verana network must respect;
- this **ECS-EGF** — the Council's own ecosystem, and deliberately the *reference implementation* of the Network GF's requirements for EGFs (chapter [NGF-EGF]); and
- the **Template EGF** — a voluntary scaffold for third parties authoring sector EGFs, for which this document serves as the worked example.

This EGF governs **only** the ECS ecosystem. It creates no obligation for any other ecosystem, and the Council is expressly not a sector-EGF authority. The Council acts here in the role the VPR specification calls an ecosystem's **governance authority**: responsible for enforcing the rules of this framework and, when necessary, applying the financial sanctions the protocol provides.

The objective of this framework, in one sentence: **the trustworthiness of the four ECS credential chains** — that a credential which verifies against the ECS ecosystem's trust root was issued by an accredited party, about a verified subject, under published rules, and that its revocation state is reliable.

1.4 How to read this document

Chapters 2–3 define conformance language and local terminology. Chapters 4–5 establish the ecosystem's on-chain footprint and the normative configuration of the four schemas. Chapter 6 contains one credential framework per schema, on a fixed skeleton. Chapters 7–8 define the accreditation machine and the Participant lifecycle. Chapters 9–14 cover economics, trust assurance, sanctions and disputes, privacy, amendment, and transition. Annex A collects every quantified threshold as an Initial Parameter; Annex B schedules this framework's Controlled Documents.

Design decisions not settled by the specifications or the Council's public record are marked with **[DECISION]** blockquotes for the Founding Members to confirm or amend.

2. Conformance

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC 2119 and RFC 8174 when, and only when, they appear in all capitals, as shown here.

Every normative statement in this document carries a stable bracketed requirement identifier (for example [ECS-ACC-4]). Identifiers are permanent: once assigned they are never renumbered or reused, and cross-references cite identifiers, never section numbers. Statements not marked normative, and all text under a heading annotated "This section is non-normative.", are informative.

Cross-references to the Network Governance Framework cite its chapter identifier prefixes (for example "chapter [NGF-EGF]"), because the sibling documents are versioned independently.

Bracketed anchors of the Verifiable Trust and VPR specifications (for example [ECS-SERVICE], [ECS-ORG], [ECS-PERSONA], [ECS-UA], or [MOD-PP-MSG-3]) are external citations into those specifications. They are distinct from this document's own requirement identifiers, whose families always carry a numeric suffix (for example [ECS-ORG-14] is a requirement of this document; [ECS-ORG] is the VT specification's Organization schema definition).

3. Terminology

Terms of art defined in the Verifiable Trust specification and the VPR specification are used with their spec meanings and are not redefined here — in particular: Corporation, Ecosystem, Participant, onboarding process, CredentialSchema, SchemaAuthorizationPolicy, trust deposit, trust unit (TU), the participant roles (ECOSYSTEM, ISSUER_GRANTOR, VERIFIER_GRANTOR, ISSUER, VERIFIER, HOLDER), and the onboarding modes (OPEN, GRANTOR_ONBOARDING_PROCESS, ECOSYSTEM_ONBOARDING_PROCESS, ISSUER_ONBOARDING_PROCESS, PERMISSIONLESS). Definitions below contain no normative keywords.

- **Accreditation** — the formal recognition, by the ecosystem governance authority or an Issuer/Verifier Grantor, of a party's competence to act in a permissioned role (issuer, verifier, grantor) for a given schema. Distinct from *certification*, which attests conformity of a product, process, or service (the UserAgent framework is a certification at software-product-line level).
- **Cause** — any of the enumerated grounds for sanction listed in [ECS-SANC-2].
- **Controlled Document** — a document subordinate to this framework, independently versioned, listed in Annex B with its owner and normative location, and binding by incorporation.
- **Council** — the Verana Council Association (in formation), a Swiss Verein (Art. 60 ZGB); in this document, always acting in its capacity as the ecosystem governance authority (EGA) of the ECS ecosystem.
- **ECS whitelist** — the list of trusted ECS Ecosystem DIDs that each VS and VUA operator maintains per the Verifiable Trust specification ([WL]).
- **Ecosystem governance authority (EGA)** — the party responsible for developing, publishing, maintaining, and enforcing an ecosystem's governance framework. Never abbreviated "GA" in this document.
- **Evidence Checklist** — the uniform, per-role, per-schema list of evidence items an applicant must satisfy; a Controlled Document whose completed instance is digest-anchored on-chain (see [ECS-ACC-8]).
- **General Assembly** — the supreme organ of the Verana Council Association under its statutes. Always spelled out; never "GA".
- **Grace Period** — the defined transition window after an involuntary termination of an issuer during which credentials it issued continue to verify while holders migrate (see [ECS-LIFE-16]).
- **Member** — an organization holding a seat in the Verana Council Association (a Verein member); per the Network GF ([NGF-TRAN-7]), the "Founding" designation is historical only and confers no right or power. Never used for a VPR role; on-chain roles are always called *Participants*.
- **Network validator** — an operator of a consensus node of the Verana network (a Council member duty). Distinct from an *onboarding validator*.
- **Onboarding validator** — the Participant that validates an applicant in a VPR onboarding process (the *validator* of the VPR specification's Participant module). The bare word "validator" is never used alone in this document.
- **Participant Candidate Agreement** — the uniform agreement executed by every applicant for a permissioned role in the ECS ecosystem; a Controlled Document derived from this framework (Annex B).

4. The ECS Ecosystem on the VPR

4.1 Governance authority and relationship to the Network GF

[ECS-GEN-1] This framework supplements, and does not supersede, the Network Governance Framework; in the event of conflict between this framework and the Network Governance Framework, the Network Governance Framework SHALL control.

[ECS-GEN-2] This framework MUST satisfy every requirement that Network GF chapter [NGF-EGF] imposes on ecosystem governance frameworks anchored on the Verana network.

Non-normative note: the ECS-EGF is deliberately the **reference implementation** of chapter [NGF-EGF]: the Council holds its own ecosystem to the same rules it asks every other ecosystem to meet, and to nothing less. Table 1 maps each [NGF-EGF] requirement class to the section of this document that satisfies it.

Table 1 below is a non-normative index; the cited sections are the normative text.

Table 1 — Conformance mapping to Network GF chapter [NGF-EGF]:

[NGF-EGF] requirement class	Satisfied by
Published on-chain as GovernanceFrameworkVersion + GovernanceFrameworkDocument (url + digest_sri, language)	§4.2, [ECS-GEN-6]
Document-control metadata block	Front-matter table of this document
Identified EGA with contact point	§4.1, [ECS-GEN-3]; contact per [ECS-GEN-4]
Legal jurisdiction stated	[ECS-GEN-5]
No contradiction of the Network GF	[ECS-GEN-1]
Layering clause	[ECS-GEN-1]
Scope defined	§1.3, §5
Roles per schema defined	Chapter 6 (per-schema frameworks)
Onboarding requirements per role	Chapters 6–7
Sanctions and appeal	Chapter 11
Issuer termination duties	[ECS-LIFE-18]
Verified revocation requests	[ECS-SANC-9]
Revocation-availability asymmetry	[ECS-LIFE-11], [ECS-GEN-7]
Fee schedule (or absence)	Chapter 9, Annex A
Privacy / data-handling commitments	Chapter 12
Amendment process with notice	Chapter 13
Personal-data prohibition on-chain	[ECS-PRIV-1]
Lawful-purpose clause	[ECS-GEN-9]
No misrepresentation of Council endorsement	[ECS-GEN-10]; §4.4 (whitelist honesty)

[ECS-GEN-3] The ecosystem governance authority of the ECS ecosystem is the Verana Council Association; the Council MUST exercise every EGA power under this framework through its own decision rules (the voting rule of Network GF [NGF-GOV-6] to [NGF-GOV-8]), and no Board organ of the association SHALL hold any authority under this framework.

[ECS-GEN-4] The Council MUST publish and maintain a public contact point for ECS ecosystem matters (currently the contact form at veranacouncil.org).

[ECS-GEN-5] The legal jurisdiction of this framework is Switzerland, the seat of the Verana Council Association; disputes escalate as provided in Chapter 11.

[ECS-GEN-6] The Council MUST anchor every ratified version of this framework on-chain as a **GovernanceFrameworkDocument** (url + digest_sri) of the ECS Ecosystem entry: version 1 is supplied to, and activated by, Create Ecosystem ([MOD-ES-MSG-1]) at Ecosystem creation; every subsequent version is added via Add Governance Framework Document ([MOD-GF-MSG-1]) and activated via Increase Active Governance Framework Version ([MOD-GF-MSG-2]); before the network is live, the canonical anchoring is the tagged release history of verana-labs/verana-council-gov.

[ECS-GEN-7] Every version of this framework and of its Controlled Documents MUST remain publicly resolvable without authentication for as long as any permission or credential issued under it can be valid; the chain-anchored digest version is canonical over any web copy.

[ECS-GEN-8] Participation in the ECS ecosystem — submitting or accepting any onboarding process, holding any Participant entry, or issuing or verifying any credential under an ECS — constitutes acceptance of the version of this framework active at the time of the act.

[ECS-GEN-9] A Participant of the ECS ecosystem MUST NOT use its permissions, or credentials issued under this framework, for an unlawful purpose.

[ECS-GEN-10] No party, including the Council, SHALL represent the creation or registration of an ecosystem, or accreditation under this framework, as an endorsement by the Council of any product, service, or organization.

[ECS-GEN-11] Nothing in this framework requires any Participant to breach law applicable to it; a Participant relying on this clause MUST notify the Council in writing, identifying the conflicting legal obligation.

4.2 On-chain footprint

[ECS-ROOT-1] The Council MUST establish and maintain the following on-chain entries on the Verana network, in this order: a Corporation entry for the Verana Council Association ([MOD-CO-MSG-1]); the ECS Ecosystem entry ([MOD-ES-MSG-1]) with its **did** (the **ECS Ecosystem DID**), carrying version 1 of this framework as its governance framework document (url + digest_sri, activated at creation; subsequent versions via [MOD-GF-MSG-1] / [MOD-GF-MSG-2]); the four ECS CredentialSchema entries ([MOD-CS-MSG-1]) configured per Chapter 5; the root **ECOSYSTEM** Participant entries ([MOD-PP-MSG-7]); and, for every (schema, role) pair with role **ISSUER** or **VERIFIER** that is subject to accreditation criteria, the corresponding SchemaAuthorizationPolicy documents ([MOD-CS-MSG-5] / [MOD-CS-MSG-6]); criteria for other gated roles are published per [ECS-ACC-1].

[ECS-ROOT-2] The Council MUST publish, in the ECS Ecosystem DID Document, one **LinkedVerifiablePresentation** service entry per ECS pointing to the self-issued Verifiable Trust Json Schema Credential (VTJSC) of that schema, with the exact fragment names required by [VT-ECS-ECOSYSTEM-DIDDOC] (**#vpr-schemas-service-vtjsc-vp**, **#vpr-schemas-org-vtjsc-vp**, **#vpr-schemas-persona-vtjsc-vp**, **#vpr-schemas-ua-vtjsc-vp**).

[ECS-ROOT-3] The VTJSCs for the Organization and Persona schemas MUST carry `validFrom` and `validUntil` properties, as required by [VT-ECS-JSON-SCHEMA-CRED-W3C]; the Council MUST re-issue each such VTJSC before its `validUntil`, without interruption of resolvability.

4.3 ECS Ecosystem DID lifecycle

Non-normative note: per [WL], VS and VUA operators whitelist the ECS Ecosystem DID as a trust root. The integrity of that single DID therefore carries the entire ECS trust graph. The following commitments are self-binding pledges by the Council, in the pattern established by root identifier governance frameworks elsewhere.

[ECS-ROOT-4] The Council MUST create the ECS Ecosystem DID and its controlling keys in a recorded ceremony involving no fewer than two authorized representatives of distinct Council member organizations, with each cryptographic action verified by a person other than the one who performed it, and MUST publish a signed ceremony report.

[ECS-ROOT-5] The Council MUST publish the ECS Ecosystem DID, and every change to it, at no fewer than three independent locations: the on-chain Ecosystem entry, `veranacouncil.org`, and the `verana-labs/verana-council-gov` repository; Council members SHOULD additionally publish it on their own websites.

[ECS-ROOT-6] The Council MUST document, as part of the ECS Onboarding and Identity Verification Operational Annex (Annex B), its key-custody, key-rotation, and key-recovery procedures for the ECS Ecosystem DID, including the multi-person control rule of [ECS-ROOT-4] applied to every rotation.

[ECS-ROOT-7] The Council MUST announce every key rotation of the ECS Ecosystem DID publicly within 7 days of execution, with the rotation's effective time and the verification path for the new key material.

[ECS-ROOT-8] If the Council ceases to operate the ECS ecosystem — by General Assembly decision, dissolution, or otherwise — it MUST, before abandoning the ECS Ecosystem DID, publish migration procedures sufficient for VS and VUA operators to re-anchor trust resolution, and MUST publicly declare the abandonment; the Council MUST NOT let the DID lapse silently.

4.4 Whitelist honesty

[ECS-ROOT-9] The Council MUST NOT represent the ECS ecosystem's trust-root status as a protocol privilege; in all public communication it MUST describe that status as recognition freely granted, and freely revocable, by individual VS and VUA operators through their ECS whitelists per the Verifiable Trust specification ([WL]).

[ECS-ROOT-10] The Council MUST NOT condition any accreditation, permission, fee, or service under this framework on a party's whitelisting of the ECS Ecosystem DID, and MUST NOT penalize any party for recognizing additional or alternative ECS ecosystems.

4.5 Trust Registry Query Protocol (TRQP) compliance

Non-normative note: the Verifiable Trust specification requires that the VPR's query API support the ToIP Trust Registry Query Protocol (TRQP). Ambiguity about what "authorized" means is the worst interoperability failure available to a trust registry, so this framework fixes the mapping for the ECS ecosystem.

[ECS-GEN-12] For TRQP queries concerning the ECS ecosystem, the PARC mapping is fixed as follows: `authority_id` = the ECS Ecosystem DID (or the Ecosystem entry `id`); `entity_id` = the `did` of the Participant entry — the DID of the Verifiable Service that appears as credential issuer or verifier (the owning Corporation is resolvable from the entry); `action` = `issue` or `verify`; `resource` = the CredentialSchema `id` of the ECS concerned; a conforming responder MUST apply this mapping.

[ECS-GEN-13] A TRQP responder answering for the ECS ecosystem MUST return `authorized: true` if and only if all of the following hold at the evaluated time: the relevant Participant entry is in `VALIDATED` state; the evaluated time falls within the entry's effective validity (`effective_from` < `t` < `effective_until`); the entry has not been revoked ([MOD-PP-MSG-9]); the Participant entry has not been ecosystem-slashed ([MOD-PP-MSG-12] — repayment per [MOD-PP-MSG-13] restores the corporation's ability to obtain a new permission, never the slashed entry); and the Participant's Corporation is not non-trustable due to an unrepaid network-level slash ([MOD-TD-MSG-5]).

[ECS-GEN-14] The Council MUST ensure that this framework is discoverable from the ECS Ecosystem DID, satisfying TRQP's requirement that the ecosystem governance framework be discoverable via the `authority_id`.

[ECS-GEN-15] Permission state for the four ECS MUST be reconstructible as-of any past time from public chain data, so that TRQP time-scoped queries and issuance-time authorization checks yield deterministic answers.

5. The Four Essential Credential Schemas

5.1 Normative configuration

[ECS-GEN-16] The Council MUST create and maintain the four ECS CredentialSchema entries with exactly the configuration of Table 2; any deviation is an amendment to this framework (Chapter 13).

Table 2 — ECS CredentialSchema configuration (normative):

Field	Service	Organization	Persona
<code>json_schema</code>	[ECS-SERVICE]	[ECS-ORG]	[ECS-PERSONA]
<code>issuer_onboarding_mode</code>	<code>ECOSYSTEM_ONBOARDING_PROCESS</code>	<code>GRANTOR_ONBOARDING_PROCESS</code>	<code>GRANTOR_ONBOARDING_PR</code>
<code>verifier_onboarding_mode</code>	<code>OPEN</code>	<code>OPEN</code>	<code>OPEN</code>
<code>holder_onboarding_mode</code>	<code>ISSUER_ONBOARDING_PROCESS</code>	<code>ISSUER_ONBOARDING_PROCESS</code>	<code>ISSUER_ONBOARDING_PRO</code>
<code>issuer_grantor_validation_validity_period</code>	0	Annex A	Annex A
<code>verifier_grantor_validation_validity_period</code>	0	0	0
<code>issuer_validation_validity_period</code>	Annex A	Annex A	Annex A

Field	Service	Organization	Persona
<code>verifier_validation_validity_period</code>	Annex A	Annex A	Annex A
<code>holder_validation_validity_period</code>	Annex A	Annex A	Annex A
<code>pricing_asset_type</code>	TU	TU	TU
<code>pricing_asset</code>	tu	tu	tu
<code>digest_algorithm</code>	SHA384	SHA384	SHA384

Non-normative notes on Table 2:

- Every `*_validation_validity_period` is a mandatory parameter of [MOD-CS-MSG-1]; 0 is the chain value written for roles a schema does not use (the protocol reads 0 as "never expire", which is inert for a role no onboarding path reaches).
- The `holder_onboarding_mode` values for Service, Organization, and Persona are mandated by the Verifiable Trust specification ([VT-ECS-JSON-SCHEMA-VPR-CONFIG]).
- Issuer modes: Service issuers are onboarded directly by the Council (`ECOSYSTEM_ONBOARDING_PROCESS`); Organization and Persona issuers are onboarded by accredited Issuer Grantors (`GRANTOR_ONBOARDING_PROCESS`); UserAgent issuers are software vendors who self-register (`OPEN`), subject to the normative vendor duties of §6.5.

[DECISION] *Verifier onboarding mode `OPEN` for all four schemas at launch.* Rationale: verifying an ECS credential is the low-risk side of the exchange — a verifier learns only what a holder chooses to present, and gating verification would tax adoption of exactly the parties (wallets, services) the ecosystem needs. Verification fees are 0 at launch (Chapter 9), so open verifier mode creates no fee-capture risk. The alternative is `ECOSYSTEM_ONBOARDING_PROCESS` for verifiers of the Persona schema, on the argument that personal-data-bearing credentials deserve accountable verifiers; Founding Members should confirm or tighten. Tightening later is a schema-configuration amendment under Chapter 13.

A second configuration choice concerns the UserAgent holder side.

[DECISION] *UserAgent `holder_onboarding_mode` = `PERMISSIONLESS`.* The UserAgent credential is an AnonCreds credential ([VT-ECS-UA-CRED-ANON]) held by end-user agent instances; unlinkability is its design goal, and the credential is checked at reception time by the holder wallet. Creating on-chain `HOLDER` Participant entries per agent instance would build a correlatable registry of user agents — contrary to the schema's privacy purpose. The alternative (`ISSUER_ONBOARDING_PROCESS`) is rejected for that reason, but Founding Members should confirm.

5.2 Schema identity: permanent links and digests

[ECS-GEN-17] The `json_schema` of each ECS CredentialSchema entry MUST verify against the corresponding SHA-384 digest of Table 3, computed per the Verifiable Trust specification (JCS canonicalization of the schema minus `$id`).

Table 3 — ECS schema permanent links and digests (from the VT spec):

Schema	Permanent link	JCS/SHA-384 digest
ServiceCredential	https://verana-labs.github.io/verifiable-trust-spec/schemas/v4/service.json	sha384- 0v+BAFGpnBX/RVqH9dU1Mg1xMrD4AKy4qUt b1lMN4iW9I2g07XjcuFmG0f0oInP3
OrganizationCredential	https://verana-labs.github.io/verifiable-trust-spec/schemas/v4/org.json	sha384- UPn4TDQs1nMBAN3FyMzTAZ0Wp99zBjBD690jpbhw0KZj7i0rS5qPwJ2SArRz0yzyu
PersonaCredential	https://verana-labs.github.io/verifiable-trust-spec/schemas/v4/persona.json	sha384- VfXTfukS020koR5USaTfEdc4NU25m4+vNrLATnjC0r0Pn1S3tFTd0vGcFSYdjE2I
UserAgentCredential	https://verana-labs.github.io/verifiable-trust-spec/schemas/v4/ua.json	sha384- yLRK2mCokVjRlGX0nVzdEYQ1o6YWpQqgdg6+HlSxCePP+D7wvs0+70TJACLZfbF/

5.3 Schema-change policy

[ECS-GEN-18] Any change to an ECS `json_schema`, onboarding mode, digest algorithm, or pricing asset is a governance act of the Council: it MUST be approved by the Council's ⅔ supermajority, MUST be announced at least 30 days before taking effect, and MUST be reflected in an amendment of this framework whose change-log row names the on-chain acts that execute it.

[ECS-GEN-19] Adoption of a new major version of the Verifiable Trust specification's ECS definitions (with new schema digests) MUST preserve verification of credentials issued under the prior version for the remainder of their validity, unless the prior version is revoked for a documented security cause.

6. Per-Schema Credential Frameworks

Non-normative note: each of the four frameworks below follows one fixed skeleton — Purpose · Scope · Principles · Issuer policies (eligibility, identity verification, issuance, revocation, level of assurance, monitoring) · Holder policies · Verifier policies · Credential definition — so that the assurance differences between schemas are legible as deltas. Where a section states "no additional restrictions", the silence is deliberate: the general rules of this framework and the Network GF apply, and nothing more. Detailed verification ceremonies (session scripts, challenge formats, accepted registry and eID lists) live in the **ECS Onboarding and Identity Verification Operational Annex**, a versioned Controlled Document (Annex B) — not in this constitutional text.

[ECS-GEN-20] The policies of each per-schema framework in this chapter apply *in addition to* the general requirements of this framework and of the Network Governance Framework; a Participant subject to them MAY be bound by stricter obligations under a Controlled Document, and no Controlled Document SHALL loosen them.

[ECS-GEN-21] Identity verification under this chapter comprises two distinct steps that MUST both be performed: **identity assurance** (establishing that the claimed real-world subject exists and is accurately described by the evidence) and **identity authentication** (establishing that the party in the onboarding session is that subject or its authorized representative, including proof of control of the relevant DID).

6.1 Service Credential Framework

6.1.1 Purpose and scope

This section is non-normative. The Service credential identifies a Verifiable Service: its DID, name, type, description, logo, minimum-age requirement, and its terms-of-service and privacy-policy documents (each pinned by URI + SRI digest). Every VS must present a Service credential as a linked verifiable presentation, bound to an accountable entity via exactly one Organization or Persona credential (VT spec [VS-REQ]); Service-credential issuers must themselves be Verifiable Services, making the trust chain recursive.

6.1.2 Principles

This section is non-normative. The Service credential is context-independent (one credential describes one service, wherever presented) and binds the service, not its operator — operator accountability travels through the Organization or Persona credential.

6.1.3 Issuer policies

[ECS-SVC-1] **ISSUER** Participant entries for the Service schema are created exclusively by the Council through the ecosystem onboarding process (**ECOSYSTEM_ONBOARDING_PROCESS**); the Council MUST act as the onboarding validator for every Service-issuer application.

[ECS-SVC-2] An applicant for Service issuer accreditation MUST satisfy the published accreditation criteria of the active SchemaAuthorizationPolicy for (Service, ISSUER), which MUST include at least: a registered Corporation entry with demonstrated control of its DID; operation of a conformant Verifiable Service; organizational identity verified to the same standard as an Organization credential subject (§6.2.4); and sustainable financing per [ECS-FEE-6].

[ECS-SVC-3] Before issuing a Service credential, the issuer MUST validate that the applicant owns and controls the service described: proof of control of the service DID by challenge-response over the DIDComm onboarding session; resolvability and SRI-digest integrity of the declared terms-of-service, privacy-policy, and logo resources; and consistency of the declared service type and **minimumAgeRequired** with the service as operated.

[ECS-SVC-4] A Service-credential issuer MUST NOT rely, for ownership or control validation, solely on assertions or documents supplied by the applicant that are not corroborated by DID-control proof or by reliable third-party sources.

[ECS-SVC-5] A Service-credential issuer MUST revoke a Service credential within 5 days of establishing any of: loss or transfer of the subject's DID control; material misdescription of the service; a verified revocation request per [ECS-SANC-9]; or revocation of the Organization or Persona credential that binds the service to its accountable entity.

[ECS-SVC-6] Service credentials are issued at a single level of assurance, anchored to organizational identity verification per §6.2.4; future versions of this framework MAY define multiple levels.

[ECS-SVC-7] A Service-credential issuer MUST monitor, at least monthly, the continued resolvability and digest integrity of the SRI-pinned resources of credentials it has issued, and MUST open a revocation review on persistent failure.

6.1.4 Holder policies

[ECS-SVC-8] **HOLDER** entries are created by issuers through the issuer onboarding process; a holder MUST notify its issuer within 30 days of any change to the credential subject data, and immediately upon loss of DID control.

6.1.5 Verifier policies

There are no additional verifier restrictions for the Service schema (verifier mode **OPEN**; see the [DECISION] under Table 2).

6.1.6 Credential definition

[ECS-SVC-9] Service credentials MUST be W3C Verifiable Trust Credentials linked to the Service VTJSC, with subject fields exactly per the VT specification's [ECS-SERVICE] and digests computed with SHA-384.

6.2 Organization Credential Framework

6.2.1 Purpose and scope

This section is non-normative. The Organization credential binds a DID to a verified legal entity: registered name, mandatory registry identifier (**registryId**), registry URI, address, ISO 3166-1 country code, legal jurisdiction, organization kind, and an optional LEI. The subject DID is the sole authoritative identifier; everything else is descriptive evidence. This is the ecosystem's highest-assurance workhorse: it is what makes a service *accountable*.

6.2.2 Principles

This section is non-normative. Verification is registry-anchored: the authoritative source for a legal entity's existence is the public register that constitutes it, never the entity's own publications. Assurance is anchored to existing standards (eIDAS "substantial", NIST 800-63 IAL2 analogues for organizations) rather than bespoke criteria.

6.2.3 Issuer structure

[ECS-ORG-1] **ISSUER** entries for the Organization schema are created by accredited **Issuer Grantors** ([GRANTOR_ONBOARDING_PROCESS](#)); the Council **MUST** accredit Issuer Grantors per jurisdiction or region against the published Issuer Grantor criteria Controlled Document for the Organization schema ([ECS-ACC-1], Annex B), and each Issuer Grantor acts as the onboarding validator for issuer applicants within its published scope, evaluating them against the active SchemaAuthorizationPolicy for (Organization, ISSUER).

[DECISION] *Issuer Grantor scopes are non-exclusive.* More than one Issuer Grantor may be accredited for the same jurisdiction or region; exclusivity would recreate the monopoly gatekeeper this architecture exists to avoid, and the neutrality duty ([ECS-ACC-13]) plus non-discretionary admission make competition safe. The alternative (one grantor per jurisdiction) simplifies oversight; Founding Members should confirm.

6.2.4 Identity verification

[ECS-ORG-2] Before issuing an Organization credential, the issuer **MUST** verify the subject's legal existence and the accuracy of every subject attribute against at least one of the following, matching the declared [registryId](#) and [registryUri](#): (a) an extract or attestation obtained directly from the constitutive public registry of the declared jurisdiction; (b) a qualified third-party database that is periodically updated from such registries; or (c) direct official communication with the organization through contact data obtained from an independent source.

[ECS-ORG-3] Evidence found on websites, in documents, or through links provided solely by the applicant **MUST NOT** be accepted as verification evidence under [ECS-ORG-2].

[ECS-ORG-4] The issuer **MUST** verify the applicant's control of the subject DID by challenge-response within the DIDComm onboarding session, and **MUST** verify that the natural person conducting the session is an authorized representative of the legal entity.

[ECS-ORG-5] The issuer **MUST** verify consistency between [countryCode](#), [legalJurisdiction](#), and the registry evidence, and **MUST** treat a declared [lei](#) as informational unless it is corroborated by vLEI-grade evidence.

[ECS-ORG-6] The issuer **MUST** screen the subject organization against applicable sanctions lists at issuance and at each renewal, at the level of assurance stated in the Operational Annex.

[ECS-ORG-7] Identity verification for Organization credentials **MUST** meet a level of assurance at least equivalent to eIDAS "substantial" (or NIST SP 800-63 IAL2 as applied to organizational representatives), as detailed in the Operational Annex; bespoke criteria **MUST NOT** replace these anchors.

6.2.5 Issuance and revocation

[ECS-ORG-8] Each Organization credential issuance decision **MUST** involve at least two authorized persons of the issuer, separating evidence verification from issuance approval.

[ECS-ORG-9] Organization credentials **MUST** carry [validFrom](#) and [validUntil](#), with a lifetime not exceeding the Initial Parameter of Annex A; renewal requires re-verification per [ECS-ORG-2]–[ECS-ORG-7].

[ECS-ORG-10] The issuer **MUST** revoke an Organization credential within 5 days of establishing any of: dissolution or deregistration of the legal entity; loss or transfer of subject-DID control; material inaccuracy of a subject attribute; or a verified revocation request per [ECS-SANC-9].

[ECS-ORG-11] An applicant that fails Organization verification **MAY** be directed to apply for a Persona credential instead (§6.3); the issuer **MUST NOT** issue an Organization credential at a relaxed standard as a substitute.

6.2.6 Monitoring

[ECS-ORG-12] An Organization-credential issuer **MUST** retain the evidence supporting each issuance for the retention period of Annex A and **MUST** make it available to the Council on reasoned request in a sanctions or audit proceeding.

6.2.7 Holder policies

[ECS-ORG-13] A holder of an Organization credential **MUST** notify its issuer within 30 days of any change affecting a subject attribute, and immediately upon loss of DID control or commencement of dissolution or insolvency proceedings.

6.2.8 Verifier policies

There are no additional verifier restrictions for the Organization schema.

6.2.9 Credential definition

[ECS-ORG-14] Organization credentials **MUST** be W3C Verifiable Trust Credentials linked to the Organization VTJSC, with subject fields exactly per the VT specification's [ECS-ORG] and digests computed with SHA-384.

6.3 Persona Credential Framework

6.3.1 Purpose and scope

This section is non-normative. The Persona credential binds a DID to an identified natural-person controller (name, description, optional avatar, mandatory controller country code and jurisdiction). It is the accountability path for services operated by individuals — the lower-ceremony sibling of the Organization credential, with personal-data duties attached.

6.3.2 Principles

This section is non-normative. Proofing follows established person-identity standards (NIST SP 800-63A, eIDAS); recency matters more than ceremony; and the holder's privacy is protected by data minimization and the consent rules of Chapter 12.

6.3.3 Issuer structure

[ECS-PER-1] **ISSUER** entries for the Persona schema are created by accredited Issuer Grantors (**GRANTOR_ONBOARDING_PROCESS**) under the same structure, criteria publication, and neutrality duties as §6.2.3 applied to the Persona schema's criteria instruments per [ECS-ACC-1] (the (Persona, ISSUER) SchemaAuthorizationPolicy and the Persona Issuer Grantor criteria Controlled Document).

6.3.4 Identity verification

[ECS-PER-2] Before issuing a Persona credential, the issuer **MUST** perform identity proofing of the natural-person controller at a level of assurance at least equivalent to NIST SP 800-63A IAL2 or eIDAS "substantial", using government-issued identity evidence or a recognized national eID scheme as listed in the Operational Annex.

[ECS-PER-3] The identity proofing relied upon **MUST** have been performed no more than the Annex A proofing-recency period before issuance.

[ECS-PER-4] The issuer **MUST** verify the subject's control of the DID by challenge-response within the DIDComm onboarding session, binding the proofed person to the DID.

[ECS-PER-5] Where the credential subject is controlled through a guardian or legal proxy, the issuer **MUST** verify the guardianship or proxy authority itself, record it in the issuance evidence, and re-verify it at renewal.

6.3.5 Issuance and revocation

[ECS-PER-6] Persona credentials **MUST** carry **validFrom** and **validUntil**, with a lifetime not exceeding the Initial Parameter of Annex A.

[ECS-PER-7] The issuer **MUST** record the subject's consent to issuance and to any publication of subject data; withdrawal of that consent is a mandatory revocation trigger, and any published derivative of the subject data **MUST** be deleted, not merely flagged, upon revocation.

[ECS-PER-8] The issuer **MUST** revoke a Persona credential within 5 days of establishing any of: death or verified incapacity of the controller without a verified successor arrangement; loss or transfer of subject-DID control; material inaccuracy of a subject attribute; consent withdrawal; or a verified revocation request per [ECS-SANC-9].

6.3.6 Monitoring

[ECS-PER-9] A Persona-credential issuer **MUST** retain issuance evidence per the Annex A retention period, applying the data-minimization and protection duties of Chapter 12 to all personal data in that evidence.

6.3.7 Holder policies

[ECS-PER-10] A holder of a Persona credential **MUST** notify its issuer within 30 days of any change affecting a subject attribute, and immediately upon loss of DID control.

6.3.8 Verifier policies

[ECS-PER-11] A verifier of Persona credentials **MUST NOT** retain presented personal data beyond the purpose and duration for which it was presented, per [ECS-PRIV-5].

6.3.9 Credential definition

[ECS-PER-12] Persona credentials **MUST** be W3C Verifiable Trust Credentials linked to the Persona VT]SC, with subject fields exactly per the VT specification's [ECS-PERSONA] and digests computed with SHA-384.

6.4 UserAgent Credential Framework

6.4.1 Purpose and scope

This section is non-normative. The UserAgent credential lets a running user-agent instance prove which software product and version it is, without identifying its user. The issuer DID identifies the software product line; the credential is issued per instance; and the format is AnonCreds precisely so that presentations are unlinkable. This framework is a *certification of software at product-line (type) level*, not an accreditation of the instance or its user.

6.4.2 Principles

This section is non-normative. Open issuer registration keeps the wallet and agent market permissionless; accountability attaches to the vendor through its trust deposit and the vendor duties below, not through a gatekeeper.

6.4.3 Issuer policies (vendor duties)

[ECS-UA-1] **ISSUER** entries for the UserAgent schema are self-created by software vendors under **OPEN** mode ([MOD-PP-MSG-14]), subject to the vendor duties of this section, which are conditions of keeping the permission.

[ECS-UA-2] A software vendor **MUST** hold exactly one UserAgent issuer authorization per software product line, and the issuer DID of every UserAgent credential **MUST** uniquely identify that product line, per the Verifiable Trust specification ([ECS-UA]).

[ECS-UA-3] A vendor **MUST NOT** issue UserAgent credentials for software it does not produce and control, and **MUST NOT** issue under one product line's issuer DID for instances of a different product line.

[ECS-UA-4] UserAgent credentials **MUST** be issued as AnonCreds Verifiable Trust Credentials ([VT-ECS-UA-CRED-ANON]) with revocation support, and **MUST NOT** be published in any DID Document.

[ECS-UA-5] The **version** (and, where used, **build**) attributes MUST accurately identify the software running in the instance at issuance; a vendor MUST NOT attest versions it has not released.

[ECS-UA-6] A vendor MUST revoke the UserAgent credentials of instances of a release within 5 days of establishing that the release is compromised or materially misrepresents its identity, and MUST maintain the availability of its revocation registry per [ECS-LIFE-11].

6.4.4 Holder policies

There are no holder onboarding requirements for the UserAgent schema (**holder_onboarding_mode** = **PERMISSIONLESS**; see the [DECISION] under Table 2): the holder wallet enforces issuer authorization at credential reception time, per the Verifiable Trust specification.

6.4.5 Verifier policies

There are no additional verifier restrictions for the UserAgent schema.

[ECS-UA-7] A verifier of UserAgent credentials MUST NOT attempt to correlate UserAgent presentations across sessions or services.

6.4.6 Credential definition

[ECS-UA-8] UserAgent credentials MUST conform to the VT specification's [ECS-UA] with the AnonCreds credential definition referencing the UserAgent VTJSC via **relatedJsonSchemaCredentialId**, and digests where applicable computed with SHA-384.

7. Accreditation

Non-normative note: this chapter is the machine that turns published criteria into on-chain permissions. Its design premises: criteria are published before anyone applies; evaluation is separated from decision; admission is non-discretionary when criteria are met; every clock is day-counted and runs against the Council as much as against the applicant.

7.1 Published criteria

[ECS-ACC-1] For every (schema, role) pair gated by an onboarding process under Table 2, the Council MUST publish objective accreditation criteria before accepting any application for that pair: (a) for pairs with role **ISSUER** or **VERIFIER**, as SchemaAuthorizationPolicy documents on-chain (url + digest_sri, created via [MOD-CS-MSG-5] and activated via [MOD-CS-MSG-6] — the only roles the protocol accepts for such policies); (b) for pairs with role **ISSUER_GRANTOR** or **HOLDER**, as digest-anchored criteria Controlled Documents (Annex B) resolvable per [ECS-GEN-7]. References elsewhere in this framework to the published criteria of the active SchemaAuthorizationPolicy for a (schema, role) are read, for pairs under (b), as references to the corresponding criteria Controlled Document.

[ECS-ACC-2] Each SchemaAuthorizationPolicy document MUST be structured as mandate → addressed role → control practice → required evidence → verification method (on-chain state, cryptographic proof, DIDComm evidence exchange, or human review), so that an onboarding validator can evaluate applications consistently.

[ECS-ACC-15] Where the published criteria for a (schema, role) define eligibility categories (for example legal form, sector, or jurisdiction), they MUST include a catch-all category admitting applicants outside the named categories on escalated evidence, so that no legitimate applicant is structurally excluded.

[ECS-ACC-3] Where an active SchemaAuthorizationPolicy exists for a (schema, role), the granting corporation MUST issue the corresponding Issuer Authorization Credential (IAC) or Verifier Authorization Credential (VAC) to the candidate upon successful completion of the onboarding process, as the VPR specification requires.

7.2 Evaluation and decision

[ECS-ACC-4] Applications for Council-validated roles (Service issuers; Organization and Persona Issuer Grantors) MUST be evaluated by the committee the Council charters for ECS accreditation, which delivers written recommendations only; the accreditation decision itself MUST be taken by the Council.

[ECS-ACC-5] If the chartered committee is vacant or inquorate, its duties under this chapter revert to the General Assembly of the Council until the vacancy is cured.

[ECS-ACC-6] Admission is non-discretionary: the Council (or, for issuer applications, the competent Issuer Grantor) MUST accredit an applicant that meets all published criteria of the active SchemaAuthorizationPolicy, and MUST NOT apply criteria that were not published at the time the application was filed.

[ECS-ACC-7] A refusal MUST state written reasons identifying each unsatisfied published criterion, and MUST inform the applicant of the appeal path of Chapter 11; a refused applicant MAY re-apply at any time without a cooldown period.

7.3 Evidence and process discipline

[ECS-ACC-8] Each application MUST be evaluated against the uniform Evidence Checklist for its (schema, role); on validation, the onboarding validator MUST anchor the SRI digest of the completed checklist in the **op_summary_digest** field of the applicant's Participant entry when executing Set Participant OP to Validated ([MOD-PP-MSG-3]), making the evidentiary basis of every accreditation verifiable.

[ECS-ACC-9] Every applicant for the same (schema, role) MUST execute the same Participant Candidate Agreement (Annex B), identical in substance for all applicants of that role; side agreements altering accreditation conditions are void as against this framework.

[ECS-ACC-10] Applications MUST be processed in the order received (FIFO) within each (schema, role) queue; the Council MAY cap concurrent substantive reviews at the Annex A workload limit, and MUST publish the current queue state.

[ECS-ACC-11] The following clocks apply symmetrically and are Initial Parameters (Annex A): the applicant completes its Evidence Checklist within the evidence window; the reviewing body completes its completeness check within the completeness-check period and its substantive review within the substantive-review period; an application with no decision at the decision deadline escalates automatically to the Council plenary, and if still undecided 30 days later, to the General Assembly.

[ECS-ACC-12] An accreditation applicant that is a Council member, or a related party of one, MUST be evaluated and decided by a panel of disinterested members per [ECS-SANC-6] applied by analogy.

7.4 Continuing duties and neutrality

[ECS-ACC-13] A fee-earning grantor or onboarding validator (an Issuer Grantor, or the Council when collecting validation fees) **MUST** evaluate every applicant only against the published criteria of the active SchemaAuthorizationPolicy, **MUST NOT** refuse or delay a qualifying applicant on any other ground, and **MUST** retain its conformance evidence for each validation for the full validity period of that validation.

[ECS-ACC-14] An accredited Participant has a continuing duty to keep its application facts accurate: it **MUST** notify its onboarding validator of any material change within 30 days, and of any change of control within the Annex A change-of-control disclosure period.

8. Participant Lifecycle

Non-normative note: every Participant in the ECS trees moves through the same state machine, and every transition is executed by a named VPR message — the paper rule and the chain act never diverge. Stages: **Qualification** → **Application** → **Activation** → **Operation** → **Notification** → **Suspension** → **Termination** → **Transition** (the eight stage names shared across all Verana governance documents). The Participant Candidate Agreement executes within the Application stage (§8.2).

8.1 Qualification and application

[ECS-LIFE-1] Before applying, a candidate **SHOULD** self-assess against the published SchemaAuthorizationPolicy criteria and the Evidence Checklist for the targeted (schema, role); the Council **MUST** keep both publicly resolvable per [ECS-GEN-7].

[ECS-LIFE-2] An application is opened by the applicant executing Start Participant OP ([MOD-PP-MSG-1]) toward the competent onboarding validator, paying the applicable `validation_fees` and the protocol's trust-deposit increment; the DIDComm validation session that follows is the venue for identity verification (Chapter 6) and evidence evaluation (Chapter 7).

[ECS-LIFE-3] An applicant **MAY** cancel its pending onboarding process at any time via Cancel Participant OP Last Request ([MOD-PP-MSG-6]).

8.2 Agreement and activation

[ECS-LIFE-4] Before activation, the applicant **MUST** have executed the Participant Candidate Agreement for its role; the agreement's confidentiality and cooperation clauses bind from signature, and full participant status attaches only at activation.

[ECS-LIFE-5] Activation is the onboarding validator's execution of Set Participant OP to Validated ([MOD-PP-MSG-3]), setting the `VALIDATED` state with `effective_from` / `effective_until` per the schema's validity periods, anchoring the completed Evidence Checklist digest per [ECS-ACC-8], and — where an active SchemaAuthorizationPolicy exists — accompanied by IAC/VAC issuance per [ECS-ACC-3]; the on-chain `VALIDATED` state is the constitutive act of accreditation.

[ECS-LIFE-6] An onboarding process that has not reached `VALIDATED` state within the Annex A activation window lapses: the onboarding validator **MUST** close it, and the applicant may re-apply without prejudice.

8.3 Operation

[ECS-LIFE-7] A validated Participant **MUST** operate in conformance with this framework, its Participant Candidate Agreement, and the SchemaAuthorizationPolicy under which it was accredited, for the entire validity of its permission.

[ECS-LIFE-8] A Participant whose validation is due to expire **MUST** renew it via Renew Participant OP ([MOD-PP-MSG-2]) with its existing onboarding validator, which re-verifies the deltas against the then-active criteria; renewal is only possible with the same, still-active validator, so if the Participant's onboarding validator has itself been revoked, the Participant **MUST** obtain a new permission by starting a new onboarding process ([MOD-PP-MSG-1]) with another competent onboarding validator.

[ECS-LIFE-9] Requalification at renewal is conducted under the criteria in effect at renewal time, not those of the original accreditation.

[ECS-LIFE-10] The Council owes decision-service levels symmetric to those it imposes: renewal reviews are subject to the same clocks as initial applications ([ECS-ACC-11]), and a renewal filed before expiry that remains undecided at expiry **MUST NOT** lapse by reason of the reviewing body's delay alone.

[ECS-LIFE-11] Availability asymmetry: a Participant's issuance operations **MAY** degrade or pause, but its revocation capability, revocation-registry availability, and the accuracy of its on-chain permission state **MUST NOT** be allowed to degrade; a Participant unable to honor this **MUST** notify per [ECS-LIFE-12] and request suspension.

8.4 Notification

[ECS-LIFE-12] A Participant **MUST** notify its onboarding validator (and the Council, for Council-validated roles) of: any breach of this framework or security incident affecting its duties, within 72 hours of discovery; any compromise of key material used in its ECS role, within 24 hours; any change of control, within the Annex A disclosure period; and any insolvency or dissolution event, immediately.

[ECS-LIFE-13] The Council owes participants no less: it **MUST** give at least 30 days' notice of material changes to this framework or its Controlled Documents (except emergency security measures, which require prompt notice and after-the-fact ratification per Chapter 13).

8.5 Suspension

[ECS-LIFE-14] A suspension is executed on-chain by the Participant's onboarding validator shortening the Participant's `effective_until` via Set Participant Effective Until ([MOD-PP-MSG-8]), after the due process of Chapter 11 — the protocol permits only the direct onboarding validator to execute this act on an OP-managed entry, so where the Council is not the direct onboarding validator it **MUST** direct the competent Issuer Grantor to execute the shortening and, failing cooperation, **MAY** proceed to revocation ([MOD-PP-MSG-9]), which the Council as ecosystem controller can always execute; a suspension notice **MUST** state the grounds by requirement identifier, the cure conditions, and the review path.

[ECS-LIFE-15] No suspension may be indefinite: a suspension review, on the Participant's request, **MUST** conclude with one of exactly three outcomes — reinstatement, continued suspension with stated cure conditions, or termination — and a suspension unresolved after the Annex A auto-termination period terminates the Participant automatically. Reinstatement of a previously suspended Participant **MUST** be recorded on the public record together with the fact and duration of the suspension.

8.6 Termination

[ECS-LIFE-16] Termination of a permission is executed by Revoke Participant ([MOD-PP-MSG-9]) by an authorized actor (an ancestor onboarding validator in the branch, the grantee itself, or the Council as ecosystem controller); an involuntary termination of an issuer opens the Grace Period of Annex A, during which: the terminated issuer's permission is dead immediately ("write death first" — it MUST NOT issue), credentials it issued continue to verify, holders migrate to a surviving issuer, and at the end of the Grace Period the issuer's unexpired credentials are revoked.

[ECS-LIFE-17] Where a termination is grounded in fraud or in damage quantifiable against the Participant's trust deposit, the Council or the competent ancestor onboarding validator MAY additionally execute Slash Participant Trust Deposit ([MOD-PP-MSG-12]), under the constraints of Chapter 11; slashed amounts are burned by the protocol ([MOD-TD-MSG-7]), and the slashed Participant may return only after Repay Participant Slashed Trust Deposit ([MOD-PP-MSG-13]) and fresh accreditation.

[ECS-LIFE-18] A terminating issuer (voluntary or involuntary) MUST: notify every affected holder of the termination and of the migration deadline within 10 days of the termination act; refrain from any new issuance from the moment of termination; revoke its unexpired credentials no later than the end of the Grace Period; archive its issuance evidence per the Annex A retention period and hand it over to the Council or a successor issuer on request; and destroy its issuance key material after the Grace Period, attesting destruction to the Council.

[ECS-LIFE-19] If a terminating issuer defaults on its notification duty, the Council MUST take over notification of affected holders and MAY hold the defaulting issuer liable for the cost.

[ECS-LIFE-20] An issuer MUST NOT transfer its holder portfolio to another issuer on its own initiative; migration is each holder's choice among accredited issuers.

8.7 Transition

[ECS-LIFE-21] A voluntary exit requires 30 days' written notice to the Participant's onboarding validator and triggers the duties of [ECS-LIFE-18]; the public record MUST state the nature of every departure of an accredited Participant (voluntary exit, expiry without renewal, or termination with its ground), symmetrically with accreditations.

9. Fees and Economics

Non-normative note: the ECS ecosystem's economics are protocol mechanics, not commercial terms. Fees are denominated in trust units (TU) per the schema configuration of Table 2 and recorded per Participant entry ([validation_fees](#), [issuance_fees](#), [verification_fees](#)); trust deposits accrue and behave exactly as the VPR specification defines — every increment is computed as the [trust_deposit_rate](#) share of the trust fees of the operation, so the fee values this chapter sets directly determine how much deposit accrues.

9.1 Launch fee schedule

[ECS-FEE-1] The [validation_fees](#), [issuance_fees](#), and [verification_fees](#) for every Council-held Participant entry in the four ECS trees are set to the Annex A values (0 TU at launch); a Participant other than the Council MUST NOT charge fees for ECS operations except per [ECS-FEE-5].

[DECISION] *Launch values of 0 TU for validation, issuance, and verification fees on all four schemas.* Rationale: the ECS are the bootstrap layer of the entire network — every Verifiable Service needs them before it can do anything else — and pricing the on-ramp would tax adoption exactly when the network needs participants most. Disclosed consequence: trust-deposit increments are computed by protocol as the [trust_deposit_rate](#) share of the trust fees of each operation, so at 0 TU fees **no ECS trust deposit accrues** — launch-cohort Participants carry no slashable ECS deposit, and the financially-backed rung (f) of the sanctions ladder ([ECS-SANC-3]) is inoperative until non-zero fees activate. The alternative — small non-zero fees from day one to fund grantor operations and build deposit collateral — is viable later via the Annex A change process and [ECS-FEE-5]; Founding Members should confirm the zero-fee launch with this trade-off in view.

[ECS-FEE-2] Trust-deposit mechanics are protocol-defined: where fees are non-zero, every applicant and Participant MUST fund the trust-deposit increments the VPR protocol requires (the [trust_deposit_rate](#) share of the applicable trust fees), and the Council MUST NOT promise their return (trust deposits are non-withdrawable by design); at the zero-fee launch schedule no ECS trust-deposit increment accrues.

9.2 Economic firewall

[ECS-FEE-3] The Council MUST NOT contribute funds of any form to the operations of any ECS Participant — no grants, subsidies, fee waivers targeted at individual parties, loans, or guarantees; this operationalizes the Council's hard limit against grant-making.

[ECS-FEE-4] The Council MUST NOT derive discretionary revenue from ECS sanctions: slashed deposits are burned by the protocol, and no fee or penalty under this framework may be routed to the Council beyond the protocol-defined flows.

9.3 Fee autonomy of grantors and issuers

[ECS-FEE-5] If and when non-zero fees are activated by amendment of Annex A, an Issuer Grantor or issuer MAY set its own [validation_fees](#), [issuance_fees](#), and [verification_fees](#) within the published bounds of Annex A, MUST publish its fee values, MUST apply them uniformly to all applicants of the same (schema, role), and MUST give 30 days' notice of increases.

[ECS-FEE-6] An applicant for a fee-earning accredited role MUST demonstrate sustainable financing of its role operations as part of its Evidence Checklist, and MUST reaffirm it at each renewal.

[ECS-FEE-7] No Participant may impose contractual lock-in or exclusivity on holders or downstream applicants; a holder MAY migrate to any accredited issuer at any time.

Non-normative honesty note: protocol switching costs exist even without contractual lock-in — where fees are non-zero, a fresh onboarding process means new trust-deposit increments, and deposits are non-withdrawable. This framework bans adding contractual friction on top; it does not pretend the protocol friction away.

10. Trust Assurance

This chapter adds no obligations for Participants. It defines how conformance with the obligations stated elsewhere in this framework is asserted, evidenced, and evaluated; its normative statements bind only the Council's own assurance practice.

10.1 Mechanism classes

[ECS-TA-1] Assurance mechanisms in the ECS ecosystem are classified in exactly three classes, and the Council **MUST** name the class of every assurance claim it makes or relies on under this framework: (a) **cryptographic/automatic** — enforced by the VPR protocol or by cryptography (permission state and expiry, trust deposits, slashing, digest-anchored documents, digest-anchored Evidence Checklists, issuance-time anchoring); (b) **attested** — self-certification against published criteria, reviewed before validation; (c) **assessed** — evaluation by a party other than the asserter (committee review, third-party audit).

Non-normative note: class (a) is the dominant column of the ECS assurance matrix — most **MUSTs** in this framework are machine-enforced or machine-evidenced on-chain, which is the ECS ecosystem's structural advantage over paper-only frameworks. The derived, non-normative Trust Assurance Matrix (Annex B) maps every requirement identifier of this framework to its class, responsible role, and evidence. The Matrix is deferred to a Controlled Document rather than shipped as an annex of this draft because it is derived row-by-row from the ratified requirement set: it can be frozen only once ratification freezes the identifiers, and publishing a derivative before its source invites divergence.

10.2 Assurance posture at launch

[ECS-TA-2] At launch, the required assurance level for accredited roles is self-certification: the applicant's completed, digest-anchored Evidence Checklist plus committee review before validation; the Council **MUST NOT** require third-party audits of launch-cohort applicants.

[ECS-TA-3] The Council **MUST** publish, in advance of applying them, the escalation drivers that will raise required assurance to attested or assessed levels for a (schema, role) — at minimum: aggregate credential volume, aggregate trust-deposit value at risk, and entry of regulated sectors relying on the schema — and any escalation applies only at renewal, never retroactively mid-validity.

[ECS-TA-4] On-chain state proves registration and permission status; it does not prove off-chain conduct. No party, including the Council, **SHALL** claim more than reasonable assurance from ECS mechanisms, and relying parties may infer less, but not more.

10.3 Monitoring

[ECS-TA-6] The Council **SHOULD** monitor conformance signals derivable from public chain data via trust resolvers and indexers — including revocation-registry availability, permission-state divergence, and expired-but-unrenewed permissions — and **MAY** open a review under Chapter 11 on persistent anomalies.

10.4 Honest residuals

This section is non-normative. Relying parties should know what this framework does **not** assure: (1) **revocation latency** — a verifier caching credential or permission state may honor a revoked credential until its cache refreshes; (2) **whitelist centralization** — trust-root status is a social convention of VS/VUA operators ([WL]), and a widely shared default whitelist is a concentration risk this framework can acknowledge but not remove; (3) **slashes are burned, not compensating** — a damaged party gains accountability from a slash, never restitution; (4) **identity-verification quality** — verification is performed by accredited humans and processes and is assured by classes (b) and (c), not by the protocol. The Council answers for the efficacy of these rules, not for the conduct of every Participant; accreditation is not a conduct guarantee.

11. Sanctions and Disputes

11.1 Scope and grounds

[ECS-SANC-1] This chapter governs sanctions at ECS-ecosystem level only; network-level sanctions (including network slashing via governance proposal) are governed exclusively by Network GF chapters [NGF-SANC] and [NGF-DISP], and nothing here reaches into any other ecosystem.

[ECS-SANC-2] Sanctions may be grounded only in enumerated **Cause**: material breach of this framework or a Controlled Document binding the Participant; material breach of the Participant Candidate Agreement; material inaccuracy in accreditation evidence or breach of the continuing accuracy duty ([ECS-ACC-14]); breach of the neutrality duty ([ECS-ACC-13]); failure of the availability asymmetry ([ECS-LIFE-11]); unlawful use ([ECS-GEN-9]); fraud in issuance or verification; or an unreported change of control past its disclosure deadline.

11.2 The graduated ladder

[ECS-SANC-3] Sanctions **MUST** proceed by the least severe sufficient rung of the following ladder, each rung executed only after the due process of [ECS-SANC-5]: (a) written warning; (b) remediation order with a cure period per Annex A; (c) permission restriction (conditions on issuance, or shortening of *effective_until* via [MOD-PP-MSG-8], executed by the direct onboarding validator per [ECS-LIFE-14]); (d) suspension (§8.5); (e) revocation of the permission ([MOD-PP-MSG-9]) with the Grace Period duties of §8.6; (f) ecosystem slash of the trust deposit ([MOD-PP-MSG-12], burned per [MOD-TD-MSG-7], return conditioned on [MOD-PP-MSG-13]).

[ECS-SANC-4] A slash under rung (f) **MUST** be proportionate to the gravity and gain of the violation, sized against a published schedule in the Operational Annex, and **MUST NOT** be a first-resort measure absent fraud; repeat violations of the same Cause within 12 months escalate one rung above the previous sanction.

11.3 Due process

[ECS-SANC-5] Before any on-chain sanction act, the respondent **MUST** receive: written notice stating the alleged Cause with the requirement identifiers concerned and the evidence; the applicable cure period (Annex A), except where the violation is ongoing fraud or an active security threat, in which case protective suspension **MAY** precede notice by no more than 5 days; and the identity of the deciding body.

[ECS-SANC-6] Sanction decisions **MUST** be taken by Council members that are disinterested in the matter (no related-party interest in the respondent or a direct competitor-complainant), applying an engagement filter per the Council's committee rules; if fewer than five qualifying members exist, the qualifying members recommend and the General Assembly decides.

[ECS-SANC-7] Every sanction decision **MUST** be published with written reasons on the Council's public record, symmetrically with accreditations; publication **MAY** redact personal data and active-security detail.

[ECS-SANC-8] A permission granted through a procedurally defective process is reviewable, not void: it remains effective until reviewed and ratified, amended, or revoked, so that defects do not cascade invalidity through credential chains onto innocent holders.

11.4 Revocation requests and challenges

[ECS-SANC-9] Any party MAY submit a revocation request or challenge against an ECS credential or accreditation, stating one of the enumerated grounds of the relevant framework chapter with evidence; the competent issuer or the Council MUST verify the requester's identity and the ground, and MUST respond with a reasoned decision within 15 days.

11.5 Appeals and disputes

[ECS-SANC-10] A sanctioned or refused party MAY appeal within the Annex A appeal window to a review body other than the one that decided: decisions of an Issuer Grantor appeal to the Council; decisions of the Council or its committee appeal to a panel of disinterested members not involved in the original decision; the final internal appeal lies to the General Assembly.

[ECS-SANC-11] An appeal against a slash (rung f) suspends its on-chain execution until the appeal is decided, unless the deciding body finds, in writing, an imminent-harm ground for immediate execution.

[ECS-SANC-12] Disputes between ECS Participants that do not allege Cause are resolved by: (a) good-faith bilateral negotiation; (b) failing that, mediation facilitated by the Council's chartered committee; (c) failing that, a reasoned Council decision, appealable per [ECS-SANC-10]; association-side legal disputes follow the arbitration provisions of the Council's statutes and the Network GF.

[ECS-SANC-13] Every sanction decision and every appeal decision under this chapter is subject to a decision deadline (Initial Parameter, Annex A) running from complete referral or, where the respondent is given a cure period or response window, from the respondent's complete response; a matter undecided at the deadline escalates automatically to the General Assembly, per the vacancy pattern of [ECS-ACC-5].

11.6 Public claims and marks

[ECS-TA-5] A party MAY publicly claim accreditation or conformance under this framework if and only if the claim is true as a pure function of live on-chain state per the collapse rule of [ECS-GEN-13] (VALIDATED, unexpired, unrevoked, not slashed); any such claim MUST embed or link the DID that lets a relying party resolve live status, and a party whose permission ceases to satisfy the rule MUST remove the claim within 5 days.

12. Privacy and Data Protection

[ECS-PRIV-1] No personal data SHALL be written to the Verana network in connection with the ECS ecosystem: on-chain entries carry identifiers, digests, permission state, and fee data only, and evidence lives off-chain with the party that collected it.

[ECS-PRIV-2] Every Participant processing personal data in an ECS role (notably Persona issuers and Issuer Grantors) MUST comply with the data protection law applicable to it, and, where no such law applies, MUST apply the Swiss Federal Act on Data Protection as its material minimum.

[ECS-PRIV-3] Issuance evidence MUST be limited to what the applicable Evidence Checklist requires (data minimization), retained no longer than the Annex A retention period, and protected with controls proportionate to its sensitivity.

[ECS-PRIV-4] The consent rules of [ECS-PER-7] (recorded consent, withdrawal-triggers-revocation, deletion of published derivatives) apply to every publication of personal data under this framework.

[ECS-PRIV-5] Verifiers MUST NOT retain personal data from ECS credential presentations beyond the purpose and duration for which it was presented, and MUST NOT use UserAgent presentations for cross-context correlation ([ECS-UA-7]).

Non-normative note on digests: the network anchors JCS digests of W3C credentials to establish objective issuance time. A digest is deliberately non-reversible and is anchored without the underlying data; the design intent is that anchored digests are not usable as personal data, and the UserAgent schema goes further by using AnonCreds, whose derived presentations are unlinkable by construction. The Council states this rationale here, in advance, rather than leaving the question to enforcement.

13. Amendment

[ECS-AMND-1] An amendment to this framework MUST be adopted by the Council under the voting rule of Network GF [NGF-GOV-6] to [NGF-GOV-8], after a public comment period of no fewer than 30 days on the published draft, with a published disposition of comments.

[ECS-AMND-2] Once the network is live, an adopted amendment takes legal effect only through on-chain activation of a new [GovernanceFrameworkVersion](#) ([MOD-GF-MSG-1] / [MOD-GF-MSG-2]); for material changes the version's [active_since](#) MUST lie at least 30 days after adoption, and the active version is immutable.

[ECS-AMND-3] Each row of this document's version history MUST correspond one-to-one with an on-chain version activation (once live), naming the adopting resolution.

[ECS-AMND-4] Emergency security amendments MAY bypass the comment period, with prompt public notice and ratification by the $\frac{2}{3}$ supermajority at the next Council decision cycle; an unratified emergency amendment lapses.

[ECS-AMND-5] Changes to Annex A Initial Parameters follow the change process stated per parameter, and every parameter change MUST be published with a written justification naming the trade-offs considered, together with a human-readable description of the change and its expected effect; the Council MUST review all Annex A parameters in an open public review at least annually.

[ECS-AMND-6] Controlled Documents (Annex B) amend on their own lighter track — committee proposal, 15-day public comment, Council adoption — and MUST NOT contradict this framework; on conflict this framework controls.

14. Transition

Non-normative note: this chapter states how the ECS ecosystem gets from draft to operating, with the same event-triggered honesty as the Council's own bootstrap.

[ECS-TRAN-1] The recruitment sequence for ECS Ecosystem Participants is: (a) this framework is delivered and ratified; (b) recruitment opens (target Q4 2026), in parallel with ongoing Council recruitment; (c) the initial cohort of Participants is accredited and permissioned in time for mainnet launch (target Q1 2027); the Council MUST NOT accredit any Participant before the criteria of [ECS-ACC-1] for its role are published.

[ECS-TRAN-2] Until dedicated tooling exists, candidates express interest through the veranacouncil.org contact form (inquiry type "ECS participant"); expressions of interest are non-binding, create no priority other than FIFO position once a complete application is filed, and are not published.

[ECS-TRAN-3] Pre-incorporation, 2060 OÜ as transitional steward MAY administer the candidate pipeline of [ECS-TRAN-2] (intake and due diligence preparation) but holds no accreditation power under this framework; all accreditation decisions are Council decisions per Chapter 7, and every steward function under this framework sunsets automatically at mainnet launch.

[ECS-TRAN-4] Accreditations decided before the network is live MUST be executed on-chain (Chapter 8 activation) as part of mainnet genesis or immediately thereafter, and each pre-mainnet accreditation MUST be re-verified as still meeting its criteria if more than 12 months elapse between decision and on-chain activation.

[ECS-TRAN-5] Any "founding participant" designation or visual distinction granted to the initial cohort MUST carry a published sunset date.

Annex A. Initial Parameters

This annex is normative. Every quantified threshold of this framework is collected here. Values marked "schema config" are written into the CredentialSchema entries per Table 2. The protocol caps every validity period at 3650 days (GlobalVariables).

#	Parameter	Initial value	Change process
A1	issuer_grantor_validation_validity_period (Organization, Persona)	365 days	[ECS-AMND-5] + schema config
A2	issuer_validation_validity_period (all four schemas)	365 days	[ECS-AMND-5] + schema config
A3	verifier_validation_validity_period (all four schemas)	365 days	[ECS-AMND-5] + schema config
A4	holder_validation_validity_period (Service, Organization, Persona)	365 days	[ECS-AMND-5] + schema config
A5	holder_validation_validity_period (UserAgent)	0 (no holder onboarding; PERMISSIONLESS)	[ECS-AMND-5] + schema config
A6	Credential lifetime (validUntil – validFrom): Organization, Persona, Service	≤ 365 days	[ECS-AMND-5]
A7	Credential lifetime: UserAgent	no fixed expiry; revocation-based	[ECS-AMND-5]
A8	validation_fees (all schemas, all roles)	0 TU	[ECS-AMND-5]
A9	issuance_fees (all schemas)	0 TU	[ECS-AMND-5]
A10	verification_fees (all schemas)	0 TU	[ECS-AMND-5]
A11	Fee upper bound once fees activate ([ECS-FEE-5])	0 TU until amended	[ECS-AMND-5]
A12	Grace Period after involuntary issuer termination	90 days	[ECS-AMND-5]
A13	Persona identity-proofing recency ([ECS-PER-3])	≤ 30 days before issuance	[ECS-AMND-5]
A14	Applicant evidence window ([ECS-ACC-11])	60 days	[ECS-AMND-5]
A15	Completeness-check period ([ECS-ACC-11])	10 days	[ECS-AMND-5]
A16	Substantive-review period ([ECS-ACC-11])	60 days	[ECS-AMND-5]
A17	Concurrent substantive reviews workload cap ([ECS-ACC-10])	5	[ECS-AMND-5]
A18	Activation window before an onboarding process lapses ([ECS-LIFE-6])	180 days	[ECS-AMND-5]
A19	Cure period, standard ([ECS-SANC-3] rung b)	15 days	[ECS-AMND-5]
A20	Cure period, core duties (revocation availability, permission-state accuracy, fraud-adjacent)	5 days	[ECS-AMND-5]
A21	Suspension auto-termination clock ([ECS-LIFE-15])	180 days	[ECS-AMND-5]
A22	Appeal window ([ECS-SANC-10])	30 days from notified decision	[ECS-AMND-5]
A23	Change-of-control disclosure period ([ECS-ACC-14])	30 days	[ECS-AMND-5]
A24	Incident / breach notice ([ECS-LIFE-12])	72 hours	[ECS-AMND-5]
A25	Key-compromise notice ([ECS-LIFE-12])	24 hours	[ECS-AMND-5]
A26	Notice of material framework changes ([ECS-LIFE-13])	≥ 30 days	[ECS-AMND-1]

#	Parameter	Initial value	Change process
A27	Evidence retention period ([ECS-ORG-12], [ECS-PER-9], [ECS-LIFE-18])	validity period of the supported permission + 24 months	[ECS-AMND-5]
A28	Revocation-request response deadline ([ECS-SANC-9])	15 days	[ECS-AMND-5]
A29	Key-rotation announcement deadline ([ECS-ROOT-7])	7 days	[ECS-AMND-5]
A30	Public-claim removal deadline ([ECS-TA-5])	5 days	[ECS-AMND-5]
A31	Sanction / appeal decision deadline ([ECS-SANC-13])	60 days	[ECS-AMND-5]

[DECISION] *Annex A values A1–A7, A12–A31.* The validity periods (365 days), Grace Period (90 days), proofing recency (30 days), and the process clocks are recommendations synthesized from operating precedents (GLEIF's 90-day grace period and 60/10/60-day qualification clocks; Sovrin's 180-day no-limbo rules; NIST/eIDAS proofing practice), chosen for a small launch-cohort Council. None is dictated by the specifications; Founding Members should confirm each value at ratification.

Annex B. Schedule of Controlled Documents

This annex is normative as to the list; the documents bind per [ECS-AMND-6] once adopted. Only documents the Council genuinely plans are listed; each has an owner and a target date.

Controlled Document	Charter (one sentence)	Owner	Target
Participant Candidate Agreement	The uniform agreement executed by every applicant for a permissioned ECS role, derived from this framework (replaces the abandoned participant MoU)	Council (accreditation committee drafts)	Q4 2026, before recruitment opens
ECS Onboarding and Identity Verification Operational Annex	Ceremony scripts, challenge formats, recognized registry and eID lists, sanctions-screening levels, slash schedule, and the ECS Ecosystem DID key-management procedures	Council (accreditation committee)	Q4 2026, before recruitment opens
SchemaAuthorizationPolicy documents, per gated (schema, ISSUER) and (schema, VERIFIER)	The objective accreditation criteria per ECS-ACC-1 , published on-chain (url + digest_sri) — the protocol accepts policies only for the ISSUER and VERIFIER roles	Council	With schema creation; before first application intake
Criteria Controlled Documents, per gated (schema, ISSUER_GRANTOR) and (schema, HOLDER)	The objective accreditation criteria per ECS-ACC-1 , digest-anchored and resolvable per [ECS-GEN-7]	Council	Before first application intake
Evidence Checklists, per gated (schema, role)	The uniform evidence instrument whose completed digest is anchored per [ECS-ACC-8]	Council (accreditation committee)	Q4 2026
Trust Assurance Matrix (derived, non-normative)	Mapping of every requirement identifier to assurance class, responsible role, and evidence ([ECS-TA-1])	Council	Before mainnet (Q1 2027)

Version History

Version	Date	Status	Summary
0.1	2026-07-05	Draft	First draft for co-authoring by Founding Members.